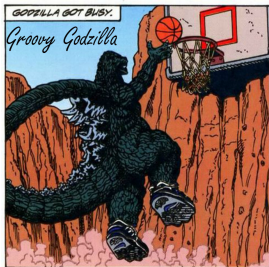


Predicting March Madness Basketball Games

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Outline

- 1 Background
- 2 Data
- 3 Past Strategies
- 4 Modeling Techniques

Section 1

Background

Introduction

- March Madness is the NCAA Division I Basketball Tournament, a single-elimination bracket of 68 teams held each March and April across both the Men's and Women's divisions to crown the National Champions.
- Teams earn a spot either by winning their conference tournament (an automatic bid) or by being selected by the NCAA Selection Committee (an at-large bid).
- The regular season determines the seeding. Teams are seeded 1 through 16 within each of four regions, where lower seeds are considered the strongest and are rewarded with matchups against the highest seeds.
- With 63 games decided by single elimination, upsets are frequent, which is why the tournament has become famous for its unpredictability, hence the name "March Madness."

Tournament Bracket

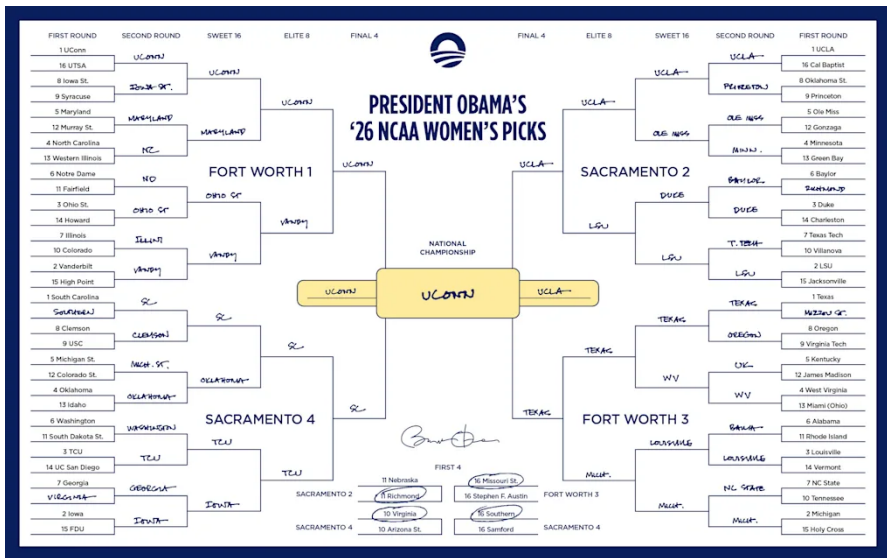


Figure 1: 2026 Bracket Example

- Kaggle is an online platform for data science and machine learning competitions, owned by Google.
- Companies, researchers, and organizations post prediction problems with datasets, evaluation metrics, and prize pools.
- Participants submit models that are scored on a hidden test set, with a public leaderboard ranking submissions in real time.
- Kaggle has become a benchmark in the ML community for evaluating predictive modeling approaches against a global field of competitors.

March Machine Learning Mania

- Annual Kaggle competition running since 2014, challenging participants to predict outcomes of every possible matchup in the NCAA tournament.
- Submissions are a probability $p \in [0, 1]$ for each of the 4556 possible matchups across Men's and Women's.
- Scored using the Brier Score:

$$BS = \frac{1}{N} \sum_{t=1}^N (f_t - o_t)^2 \quad (1)$$

- Where f_t is our prediction and o_t is the outcome.
- o_t is 1 if the team wins, and 0 if they lose.

Brier Score

- A naive 'coin flip' strategy (50% probability for all games) yields a Brier Score of 0.25, serving as our baseline.

Predicted	Outcome	Scenario	Brier Score
0.50	Win or Loss	Baseline	0.2500
1.00	Win	Perfectly confident, correct	0.0000
1.00	Loss	Perfectly confident, incorrect	1.0000
0.70	Win	Moderately confident, correct	0.0900
0.70	Loss	Moderately confident, incorrect	0.4900
0.90	Win	Highly confident, correct	0.0100
0.90	Loss	Highly confident, incorrect	0.8100

Table 1: Brier Score Examples

Prediction Difficulty

- Top predictive models plateau near 75% accuracy [?].
- Odds of a perfect bracket: 1 in 9.2 quintillion (random) to 1 in 128 billion (informed).
- No publicly verified bracket has survived past the Sweet Sixteen [?].
- \$3.1 billion wagered in 2025, sportsbook lines are the benchmark to beat [?].

Section 2

Data

Kaggle Data

- Kaggle provides 37 tables spanning over three decades, separated by gender.
- Contains detailed regular season and playoff results, plus tournament seeding tables.
- Detailed box scores begin in 2003 for men and 2010 for women.

Variable	Description
PTS	Points scored
FGM / FGA	Field goals made and attempted
FGM3 / FGA3	Three-point field goals made and attempted
FTM / FTA	Free throws made and attempted
OR / DR	Offensive and defensive rebounds
Ast	Assists
TO	Turnovers
Blk	Blocks
PF	Personal fouls

Table 2: Detailed Result Variables

External Data Sources

- Box scores miss nuances such as pace of play and strength of schedule.
- Supplement Kaggle data with advanced efficiency metrics:
 - **KenPom** (Men's): industry standard, per-possession efficiency adjusted for opponent [?].
 - **Bart Torvik** (Women's): comparable methodology, used since KenPom is men's only [?].

Metric	Description
NetRtg	ORtg – DRtg
ORtg	Pts / 100 poss.
DRtg	Pts allowed / 100 poss.
AdjT	Possessions per 40 min

Table 3: KenPom Metrics

Metric	Description
ADJOE	Adjusted offensive efficiency
ADJDE	Adjusted defensive efficiency
AdjT	Adjusted tempo

Table 4: Bart Torvik Metrics

Data Considerations

- **2020 Tournament Gap:** the COVID-canceled tournament produces no playoff observations; regular season 2020 data is still usable.
- **Completeness:** no missing values across Kaggle, KenPom, or Torvik. Every team has a complete statistical profile.
- **Team Name Matching:** external sources use inconsistent spellings (ex: “St. Mary’s” vs “Saint Mary’s CA”).
 - Stage 1: exact match on normalized names against Kaggle lookup
 - Stage 2: fuzzy match using Jaro-Winkler distance with threshold 0.3
 - All teams successfully matched in both divisions

Section 3

Past Strategies

- Tournament seeding alone is a strong predictor via logistic regression with z-score transformation [?].
- Combined logistic regression and Markov chain models, benchmarked against nearly 100 ranking systems, established logistic regression as the baseline standard [?].
- No classifier (decision trees, random forests, naive Bayes, MLPs) exceeds 74–75% accuracy on NCAAB data [?].
- **Key takeaway:** feature construction matters more than model choice.

- **Fundamental Analysis:** construct a “true” probability independent of public sentiment.
 - Elo ratings [?], KenPom, FiveThirtyEight, Massey
 - Treats expert outputs as features, not final predictions
- **Efficient Market Hypothesis [?]:** betting markets aggregate information better than any single model.
 - Uses closing lines from major bookmakers.
 - `goto_conversion` algorithm [?] corrects for favorite-long shot bias.

NIL Era and Recent Baselines

- **Boosting** [?]: tiered features (seeding to efficiency to Elo to GLM team quality), implemented in XGBoost. Multiple top 10 finishes in 2024 and 2025 cite his work as the baseline for their submission.
- **Simulation pipelines** among top 2024 finishers: predict point margins, then simulate brackets thousands of times via normal distribution over margins.
- Inspired the three models in this paper: **Logistic Regression**, **XGBoost**, and **Linear Regression Monte Carlo**.

Section 4

Modeling Techniques

Logistic Regression

- Binary classification model that applies the sigmoid function to a linear combination of features.

$$P(Y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k)}} \quad (2)$$

- $\mathbf{x} = (x_1, \dots, x_k)$: feature vector
- $\beta = (\beta_0, \dots, \beta_k)$: coefficients estimated via maximum likelihood
- Output is bounded between $[0, 1]$, making it interpretable as a probability.
- Assumes linear relationship between features and log-odds; nonlinearity addressed via polynomial or interaction terms.